

(13) num = {3, 0, 1} , ~~0~~

→ Iska XOR karo

$[3 \wedge 0 \wedge 1] \rightarrow$ Ye ^{num} ~~range~~ ka XOR.

Ab range [0, 3]

0, 1, 2, 3 → Iska XOR karo

$[0 \wedge 1 \wedge 2 \wedge 3]$

→ Ab in dono ka XOR karo

$[3 \wedge 0 \wedge 1] \wedge [0 \wedge 1 \wedge 2 \wedge 3]$

XOR properties.

$$\begin{cases} a \wedge 0 = a \\ a \wedge a = 0 \end{cases}$$

To,

$$0 \wedge 0 = 0$$

$$1 \wedge 1 = 0$$

$$3 \wedge 3 = 0$$

In teens me bacha
leiy a = 2.

→ Ye ho jayega
missing number.

Note:-

- Jo num given uske element me XOR karo.
- Phir jo range hai uske each element ka XOR karo.
- At the end num or range ke element ka XOR karo. missing number mil jayega.

int temp = 0;

for (int i = 0; i < nums.length; i++) {
 temp = temp ^ i;

}
for (int i = 0; i < nums.length; i++) {
 int num = nums[i];
 temp = temp ^ num;

}
return temp;

→ Is loop me hm range ke sare element ka XOR karenge.

→ Aur Is loop me given nums ke element ka XOR karenge.